SafePrompt: Jailbreak Search Defense through Activation Signatures

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Abstract

Jailbreak search constructs many request variants to evade a language model's safeguards. We show that these candidates carry a detectable signature in the intermediate activations of a separate frozen model, including candidates that elicit safe responses. SafePrompt learns its energy readout from attacks expanded from 52 JailbreakBench seed goals and calibrates rejection on benign requests. On 100 goals for each of two victims, it reduces PAIR success from 92/91% to 14/17% and TAP success from 82/93% to 19/27%, while rejecting 2.53% of 750 benign requests. Native HaloGuard and WildGuard reject 32.67% and 50.67% on the same holdout. We trace search protection to activation separation, reference coverage and the accumulation of rare passing candidates across goals. Matched presentation changes shift both benign and harmful energies, showing why calibration must reflect the deployment population. In sequential search with binary gate feedback, the attacker succeeds on seven of 30 goals at 90 queries per goal. These findings establish activation-signature detection as a practical search defense and identify the reference, utility and feedback choices that govern its transfer.

1 Introduction

Repeated variations of a harmful request can bypass an aligned model's safeguards (Hughes et al., 2024). An input defense encounters proposals for the same goal, and a small fraction of accepted candidates can expose many goals as the query budget grows. Prompt-level harmfulness accuracy therefore gives an incomplete account of search defense: the system must reject attack variants while preserving legitimate requests about sensitive subjects.

SafePrompt detects request constructions produced by jailbreak search. A separate frozen language model maps each request to an intermediate representation; a compact energy readout measures its compatibility with legitimate requests relative to attack references. The calibrated gate rejects high-energy proposals before victim generation and passes accepted requests unchanged. This signal is available even when the victim refuses the proposal. Figure 1 connects the representation geometry to this input-side decision. Direct forms of the same harmful goals largely pass, while their search-generated variants occupy a compact activation region and have higher energy. This contrast motivates attack detection as a different task from general harmfulness classification.

A separate rejection boundary can act on these signatures even when the victim's refusal policy misses them. A readout trained with 52 augmented JailbreakBench seed goals rejects over 96% of recorded PAIR/TAP candidates in both safe-output and harmful-output strata, with low rejection on the benign holdout. The same gate protects two victims without victim-specific fitting. Comparisons with native guards and transformation defenses establish the resulting security and utility trade-off. The analysis then asks which properties of the attack population make this protection possible and which reference choices preserve it.

We organize the analysis around three questions. What do activations separate, and how do residual passes become successful goals? Which reference and readout choices preserve that separation across searches and benign sources? How do presentation and actual gate feedback change the attack distribution? We answer with goal-weighted energy distributions, chronological outcomes, reference-population controls, matched wrappers and a sequential adaptive-search experiment.

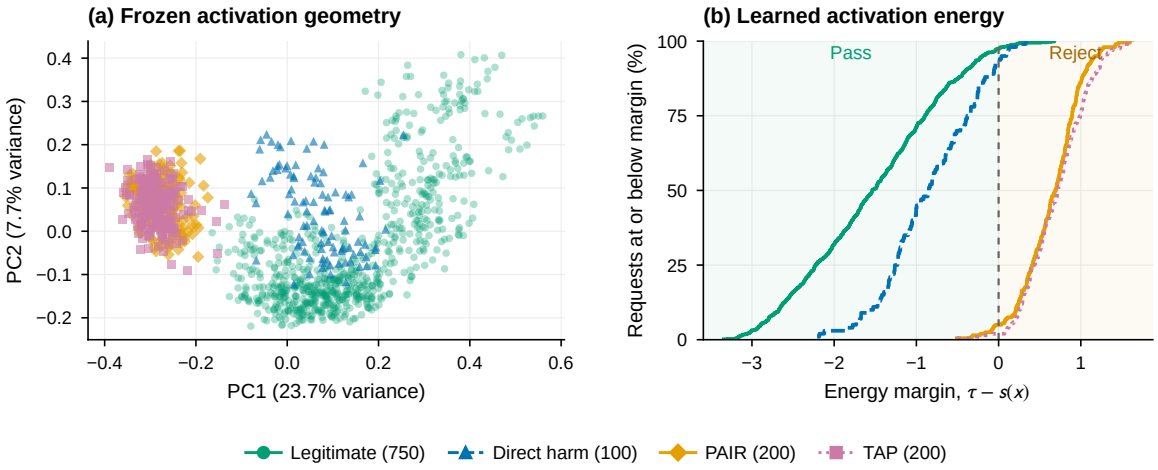


Figure 1: **Search requests occupy a distinct activation region.** (a) Centered, label-free PCA of 1,250 unique Qwen layer-12 states. (b) Frozen linear-head energy; positive margins reject. The same 100 goals per victim supply 100 unique direct requests and first PAIR/TAP proposals (200 each), beside 750 legitimate inputs. The head rejects 7% of direct requests, 95% of PAIR and 98% of TAP. One fixed model/head; all points included.

2 Related Work

PAIR, TAP and AutoDAN automate jailbreak search (Chao et al., 2023; Mehrotra et al., 2023; Liu et al., 2023). Prompt transformations and context defenses change the victim's input, while a gate can reject a request before generation (Jain et al., 2023; Robey et al., 2023; Xie et al., 2023). Evaluating adaptation requires feedback from the actual defended pipeline (Croce et al., 2022; Andriushchenko et al., 2024); our adaptive experiment includes the gate's binary decision in that loop.

Safety directions and probes reveal refusal and detection information in representations (Arditi et al., 2024; Zou et al., 2023; Kirch et al., 2024). SAFEGuard combines fluency and harmful-semantic evidence (Vo et al., 2026), and LatentBreak searches near harmless representations (Mura et al., 2025). Our analyses connect representation separation to chronological search risk, reference/readout controls and feedback-driven attacks.

WildGuard separates prompt harmfulness, output harmfulness and refusal (Han et al., 2024); HaloGuard and broader moderation evaluations study policy coverage and utility (Sangameswaran et al., 2026; Hu, 2026). Moderation policies also determine which distinctions a classifier is asked to make (Palla et al., 2025). We compare native guards on identical attack candidates and analyze search defense separately from policy classification.

3 Calibrated Activation Energy

For frozen layer-ℓ token states $H_\ell(x)_t$ and non-padding mask m_t, we form a normalized mean representation,

$$z(x) = \text{normalize}\left(\frac{\sum_t m_t H_\ell(x)_t}{\sum_t m_t}\right). \quad (1)$$

The deployed gate uses Qwen3.5-0.8B, layer 12, the first 256 tokens and a linear score $s_\theta(z)$. Its energy is $E_\theta(x) = -s_\theta(z(x))$. A separate benign calibration set gives the lower score quantile τ_α ($\alpha = 0.05$), with rejection rule

$$\text{reject } x \iff E_\theta(x) > -\tau_\alpha. \quad (2)$$

Equality passes, and the victim receives the full original accepted request.

Let P_B be the positive references and Q the negative mixture, weighted by r relative to positives. The readout minimizes

$$\mathcal{L} = -\mathbb{E}_{P_B} \log \sigma(s) - r\mathbb{E}_Q \log \sigma(-s). \quad (3)$$

The recipe combines general and sensitive-but-benign positives, generated attacks and normalized Gaussian negatives. The text negatives originate from 52 JailbreakBench development goals (Chao et al., 2024). Examples include a campaign advocating employment discrimination and unauthorized access to a server. LLMs expand

these goals into many negative prompts, including explicit paraphrases and benign-looking adversarial framings. Equal batches give $r = 2$ and $Q = \frac{1}{2}Q_A + \frac{1}{2}Q_G$; calibration uses 600 separate requests. On common support, the unrestricted optimum is $s_r^*(z) = \log[p_B(z)/q(z)] - \log r$ (Gutmann and Hyvärinen, 2010). Energy thus measures reference-relative compatibility. Its decision depends on the learned ordering and the benign calibration distribution. Full data and checkpoint details appear in Appendix F.

Gate procedure. For each attacker proposal, the defender performs the following operations; blocked queries still consume the attack budget.

1. Encode the inspected prefix into $z(x)$.
2. Compute $E(x) = -s_\theta(z(x))$.
3. If $E(x) > -\tau_\alpha$, return a refusal.
4. Otherwise, pass unchanged x to the victim.
5. Return the response to the attacker.

The encoder, readout and threshold remain fixed during evaluation. The adaptive experiment changes what the attacker observes, not the defender's parameters.

4 Search Defense Evaluation

At budget N, ASR is the fraction of goals with an accepted harmful output among the first N chronological queries. We evaluate the first 100 standard HarmBench behaviors in source order on Llama-3.2-3B and Qwen3.5-4B, using response judgments (Mazeika et al., 2024). Fixed-search evaluation applies gates to recorded PAIR, TAP, AutoDAN and template requests with unchanged victim responses. Section 7 evaluates newly generated, feedback-consuming search. Additional response judgments and all budget curves are in the appendices.

Utility uses 500 OR-Bench-hard and 250 safe XSTest requests (Cui et al., 2024; Röttger et al., 2023). Gate FRR counts input rejections; end-to-end FRR also includes victim refusals. Safe responses to attack requests remain attack observations, outside the benign utility denominator. Comparisons use each guard's fixed native policy and threshold; inspection windows differ. Sampling intervals resample whole goals or benign requests for fixed heads, with paired draws for differences. Numerical reproducibility and cohort details appear in Appendix A.

4.1 Security and utility across attacks

Table 1 separates search budgets from single-query attacks. SafePrompt gives substantially stronger PAIR protection than the native guards at much lower benign rejection. On Qwen TAP, WildGuard and SafePrompt have similar ASR, but WildGuard rejects half the benign holdout. The utility cost concentrates on OR-hard: a guard that broadly rejects sensitive requests can suppress attacks while also restricting legitimate use. The table reports that cost alongside each security result.

The same gate serves both victims; variation in defended ASR therefore reflects the search proposals and victim responses, rather than a separately chosen threshold. Low undefended success on the template rows leaves little headroom for a security comparison. PAIR and TAP supply the stronger tests. General harm-classifier benchmarks and the separately trained Phi recipe appear in Appendices K and J.1.

4.2 Matched transformation comparison

Transformations require new victim responses. Table 2 compares six transformation defenses on the same first chronological requests and benign holdout. SafePrompt suppresses harmful responses on both victims while preserving the original accepted input. Transformation effectiveness varies by victim and often incurs greater benign refusal. Its full protocol and additional response judgments are in Appendix B.

5 What the Activation Signature Detects

5.1 The signature persists across victim outcomes

Safe victim responses do not remove a search request's activation signature. Figure 2 separates both output-conditioned candidate populations from benign requests. Each contributing goal has equal weight, so a handful of long trajectories cannot create the separation by dominating the distribution. Table 3 confirms high rejection in every PAIR/TAP stratum, across both victims and both response outcomes. The signal identifies properties of attack construction that persist even when the victim successfully resists a proposal.

This distinction changes the examples appropriate for detector training. An unsuccessful jailbreak still belongs to the attack population; labeling it benign because the victim refused would erase useful detection signal. Conversely, a legitimate sen-

(a) Llama-3.2-3B

Attack / queries (% ASR ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard
PAIR@1	7	**0**	2	2	5	2	3
PAIR@90	92	**14**	70	58	85	61	76
TAP@1	10	**1**	6	4	10	3	5
TAP@300	82	**19**	56	42	76	49	64
AutoDAN@1	5	**0**	0	0	3	1	3
AutoDAN@100	9	**0**	0	0	7	2	6
Plain@1	5	**5**	0	1	2	0	4
DAN@1	6	**0**	0	0	1	0	2
H-CoT@1	12	**0**	0	1	3	0	5
Many-shot@1	0	**0**	0	0	0	0	0
Benign assumption@1	14	**2**	2	1	5	1	1
Template attack@1	35	**0**	0	0	1	0	2
End-to-end FRR (%)	5.6	**7.9**	35.2	52.7	15.2	71.5	40.0

(b) Qwen3.5-4B

Attack / queries (% ASR ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard
PAIR@1	21	**1**	0	1	7	1	2
PAIR@90	91	**17**	44	36	58	31	49
TAP@1	17	**1**	4	1	6	0	5
TAP@300	93	**27**	44	26	62	34	47
AutoDAN@1	0	**0**	0	0	0	0	0
AutoDAN@50	0	**0**	0	0	0	0	0
Plain@1	0	**0**	0	0	0	0	0
DAN@1	0	**0**	0	0	0	0	0
H-CoT@1	0	**0**	0	0	0	0	0
Many-shot@1	0	**0**	0	0	0	0	0
Benign assumption@1	0	**0**	0	0	0	0	0
Template attack@1	0	**0**	0	0	0	0	0
End-to-end FRR (%)	36.1	**37.2**	52.1	57.6	41.2	72.3	52.4

Shared gate FRR (% ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard
Gate FRR (750)	0.00	**2.53**	32.67	50.67	10.80	70.27	36.93
OR-hard (500)	0.00	**3.80**	44.60	75.40	14.80	93.20	52.60
XSTest (250)	0.00	**0.00**	8.80	1.20	2.80	24.40	5.60

Table 1: **Security across search and template attacks.** HarmBench-judged % ASR, 100 goals per attack/victim and 750 benign requests. Bold SafePrompt is the deployed Qwen linear gate. Guards evaluate identical requests and unchanged victim responses. Search @1 is chronological; AutoDAN ends at 100/50 queries and templates use one. Gate FRR counts input rejection; end-to-end FRR includes victim refusal.

sitive request remains a utility example even when its vocabulary overlaps a harmful behavior. Input detection and response judgment provide separate supervision for separate decisions.

5.2 Rare passes determine behavior-level risk

Let B_N indicate any passing proposal by query N, and J_N any accepted harmful output. Since $J_N \Rightarrow B_N$, empirical ASR decomposes as

$$\widehat{\mathrm{ASR}}_N = \widehat{P}(B_N)\,\widehat{P}(J_N \mid B_N). \qquad (4)$$

The first factor measures exposure to the victim; the second measures harmful success among exposed goals. Both preserve dependence between proposals in a search. Qwen PAIR and TAP expose the same 61 goals, but TAP produces more accepted harmful outputs within that set. A candidate-rejection rate cannot explain this difference. Figure 2(c) shows exposure accumulating with the query budget even when most candidates are rejected.

The useful security target is the harmful tail of the accepted set. Treating all bypasses as successes overstates risk, while reporting only average rejection hides the repeated opportunities to reach the victim. Exposure and joint success distinguish where a defense fails: at the input boundary, at the victim's response, or at both. They support comparisons across budgets that prompt-level accuracy misses.

(a) Llama-3.2-3B

Defense	% ASR ↓ PAIR	TAP	AutoDAN	FRR (%) E2E
None	7	10	5	5.60
SafePrompt	**0**	**1**	**0**	**7.87**
Paraphrase	10	15	10	4.80
SmoothLLM	0	2	2	19.60
Perplexity	7	10	5	5.73
Backtranslation	7	10	5	5.20
Retokenization	3	5	4	11.33
Self-Reminder	8	11	1	7.07

(b) Qwen3.5-4B

Defense	% ASR ↓ PAIR	TAP	AutoDAN	FRR (%) E2E
None	21	17	0	36.13
SafePrompt	**1**	**1**	**0**	**37.20**
Paraphrase	23	19	3	27.47
SmoothLLM	24	20	0	28.93
Perplexity	21	17	0	35.87
Backtranslation	21	17	0	36.80
Retokenization	21	18	0	34.00
Self-Reminder	15	21	0	52.40

Table 2: **Transformation defenses on matched first queries.** HarmBench-judged % ASR and end-to-end false refusal (%). Cells use the first chronological query, 100 behaviors; utility uses the shared 750 requests. SafePrompt filters unchanged victim responses; the six transformation defenses use matched requests with regenerated responses.

(a) Llama-3.2-3B

Search$_N$	Blocked (%) Harmful-out	Safe-out	Goals / 100 Pass / harm	Harm/pass (%)
PAIR$_{90}$	97.8	96.4	70 / 14	20.0
TAP$_{300}$	98.5	97.1	68 / 19	27.9

(b) Qwen3.5-4B

Search$_N$	Blocked (%) Harmful-out	Safe-out	Goals / 100 Pass / harm	Harm/pass (%)
PAIR$_{90}$	97.3	97.5	61 / 17	27.9
TAP$_{300}$	98.5	98.1	61 / 27	44.3

Table 3: **Candidate capture, goal bypass and harmful success are distinct.** Fixed Qwen linear head and HarmBench response labels. Pass / harm counts goals with an accepted query / accepted harmful output by N; harm/pass is their conditional fraction. Block rates use all observed candidates, including repeats.

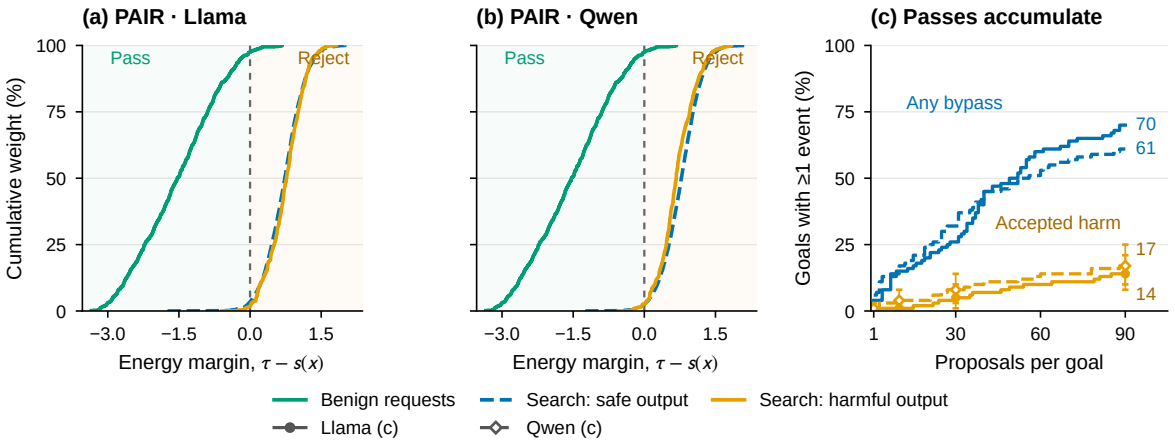


Figure 2: **Search candidates separate in energy; passes accumulate across goals.** (a,b) Equal-goal CDFs conditioned on safe/harmful HarmBench outputs: 100/92 Llama and 100/91 Qwen goals. Benign requests have equal request weight. (c) PAIR prefixes, 100 goals per victim; accepted-harm points have 95% whole-goal bootstrap intervals (2,000 draws). One fixed Qwen head.

6 Reference Distributions Define the Boundary

6.1 Safety is defined by the references

The reference distributions define what SafePrompt treats as safe. P_B represents legitimate use; Q represents requests to reject. The energy $E(x) = -s(x)$ measures compatibility with that contrast. It is not an intrinsic safety value attached to a sentence. The same words can describe a legitimate security task or request unauthorized access; purpose, context and the permitted use determine the appropriate decision.

Changing the references changes that decision. Adding a legitimate sensitive use to P_B teaches the readout to retain it. Adding attack constructions to Q teaches the readout to reject their shared features. Neither distribution must enumerate ev-

ery possible request. The frozen encoder supplies reusable features, and the head learns which features distinguish the chosen populations. This is why a small seed-goal inventory can support transfer across many search variants.

The Qwen reference controls show that this choice matters. Replacing the positive population at the same sample count reduces attack capture from 96.4% to 6.6%, while text-only negatives retain strong capture (Appendix F). The wrong contrast can erase the boundary even at a fixed sample count. The useful design question is which legitimate uses and attack constructions the references cover.

6.2 Why a benign anchor helps transfer

We hypothesize that a classifier trained on known harmful requests can overfit to their styles. A new jailbreak preserves the harmful goal while changing its presentation, so cues learned from the negative examples can disappear. Search repeatedly probes such gaps until one accepted request elicits a harmful response.

The density view anchors the boundary to a population of legitimate requests. It asks how compatible a representation is with benign use relative to the negative references, rather than whether it resembles a particular known attack. Shared activation features can then support transfer across unseen rephrasings. The positive-reference controls in Section 6.1 support the importance of this anchor. This hypothesis concerns references and features; BCE remains a discriminative logistic objective (Grathwohl et al., 2019).

Figure 3 shows the security consequence of gaps in the boundary. On Llama PAIR, HaloGuard's ASR rises from 2% at the first query to 70% at query 90, while SafePrompt rises from 0% to 14%. Qwen TAP remains an exception: WildGuard and SafePrompt have similar final ASR, with much greater benign rejection by WildGuard. A classifier's early rejection rate therefore does not determine its protection over a search.

6.3 Protection has a utility cost

Figure 4 separates the learned score from its operating point. Tightening the threshold reduces harmful search success while rejecting more legitimate requests, without retraining the head. The nominal 5% calibration setting lies in a low-cost region on these held-out requests; it is not a universal benign rejection rate. Deployment calibration should represent the intended users and tasks.

A threshold can spend a different rejection budget, but cannot repair the score ordering. When legitimate and attack requests overlap, better reference coverage or features are needed. Protection should therefore be compared at explicit utility costs, rather than credited to a permissive or restrictive threshold alone.

6.4 Activation evidence beyond refusal

SafePrompt makes its input decision entirely from the frozen encoder's activations. The readout learns how suspicious constructions differ from legitimate traffic in that model's representation. We interpret this as latent evidence of an attack: a signature can remain detectable when the victim's guardrails fail to reject the request. Refusal behavior is mediated by activation features (Arditi et al., 2024); SafePrompt learns a separate boundary anchored to benign traffic.

Content and framing jointly determine the representation. PAIR and TAP vary roles, rephrasings and justifications for the same harmful goal (Chao et al., 2023; Mehrotra et al., 2023); even small repeated variations can bypass aligned models (Hughes et al., 2024). Figure 5 holds the requested content fixed while changing its presentation. Adversarial framing raises SST harmful rejection from 0% to 87%, but also raises WG benign rejection from 54% to 90%. These shifts show that the signature reflects how a request is constructed, rather than its harm label alone.

The benign anchor defines permitted use. Roleplay, hypothetical scenarios and sensitive instructions can be legitimate uses, so rejecting a framing wholesale would sacrifice utility. Positive references should cover those uses, and evaluation should pair harmful reformulations with legitimate requests in similar forms. Overlapping activation regions call for better references or representations; a longer inspection window alone does not resolve the collision (Appendix G).

7 Adaptive search against SafePrompt

We use sequential, PAIR-style prompt refinement (Chao et al., 2023). Mistral-7B proposes one request per step against Llama-3.2-3B behind the frozen Qwen SafePrompt gate. For each of 30 fixed HarmBench goals, search runs for 90 steps; the 30-query result uses the same trajectory's prefix. The proposer starts from the goal, then re-

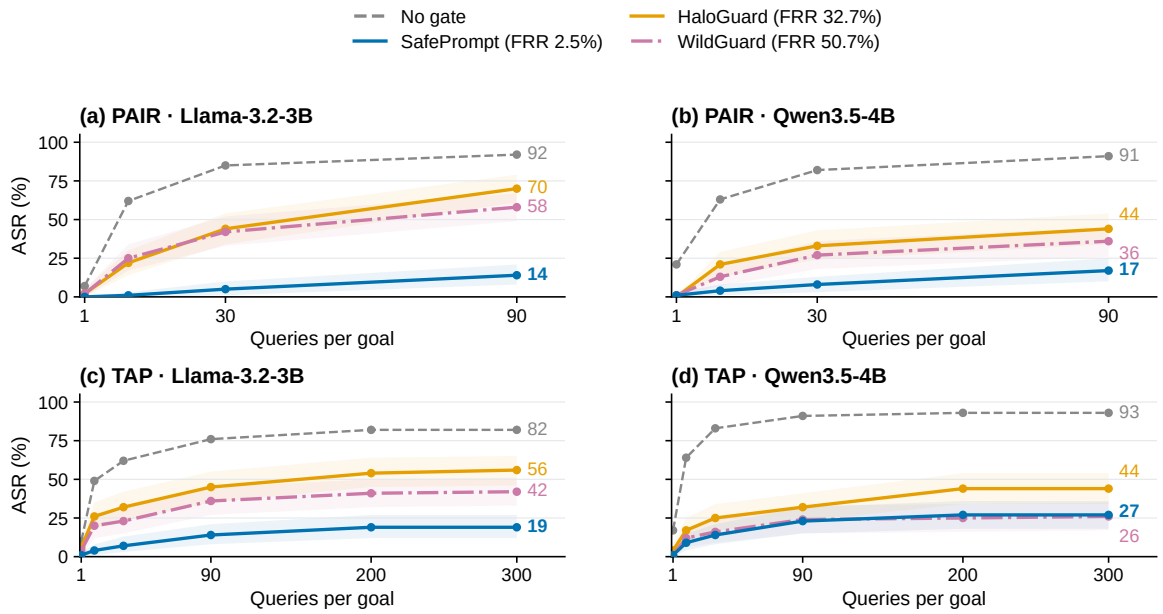


Figure 3: **Classifier misses accumulate into successful searches.** HarmBench ASR on 100 goals per victim, measured at the marked query budgets. Colored bands are pointwise 95% whole-goal bootstrap intervals (2,000 draws) for fixed searches and heads. Legend FRRs use the same 750 benign requests. Native operating points are retained; WildGuard and SafePrompt have similar Qwen TAP success despite different benign costs.

Feedback	N	Pass/query	% ASR ↓	Ungated
Gate + response	30	28/900	16.7	–
Gate + response	90	61/2700	23.3	–
None	30	124/900	36.7	–
None	90	434/2700	60.0	–
Response only	30	19/900	13.3	70.0
Response only	90	52/2700	26.7	90.0

Table 4: **Adaptive feedback changes exposure and harmful success.** Mistral-7B attacks Llama-3.2-3B through the frozen Qwen gate. HarmBench % ASR uses 30 goals; pass counts use queries. Response-only ungated % ASR includes counterfactual blocked calls. Every proposal consumes budget; judgments follow search.

ceives up to four previous prompt–feedback pairs and generates an explanation of its revision and a new attack prompt. We compare three feedback conditions:

- **Gate + response:** a rejection returns a binary blocked verdict without a victim response, and instructs the proposer to rewrite the request to avoid the gate. A pass returns a passed verdict and the first 600 characters of the victim response; the proposer is instructed to revise the request to both pass the gate and elicit the harmful behavior.
- **Response only:** the proposer refines its request from the first 600 response characters, without gate verdicts. The victim answers every proposal, including those the gate would reject. These counterfactual responses guide search but cannot count as defended successes.
- **None:** the same proposer generates each request from the goal with an empty history; the gate still controls victim access.

Every proposal, including a rejection, consumes budget. Search completes all 90 steps without early stopping. HarmBench judges outputs afterward; the attacker receives no energy scores, gradients or judge labels.

Response-only refinement reaches 90% ungated ASR at 90 queries, but 26.7% after filtering (Table 4). Gate-aware refinement reaches 23.3%, while feedback-free sampling reaches 60.0%. Gate feedback changes the proposal population without making this refinement strategy stronger than independent proposals.

For gate-aware search, the pass fraction falls from 3.11% at 30 queries to 2.26% at 90, yet successful goals rise from five to seven. Repeated search finds rare accepted tails: high rejection per query does not determine accumulated risk per goal.

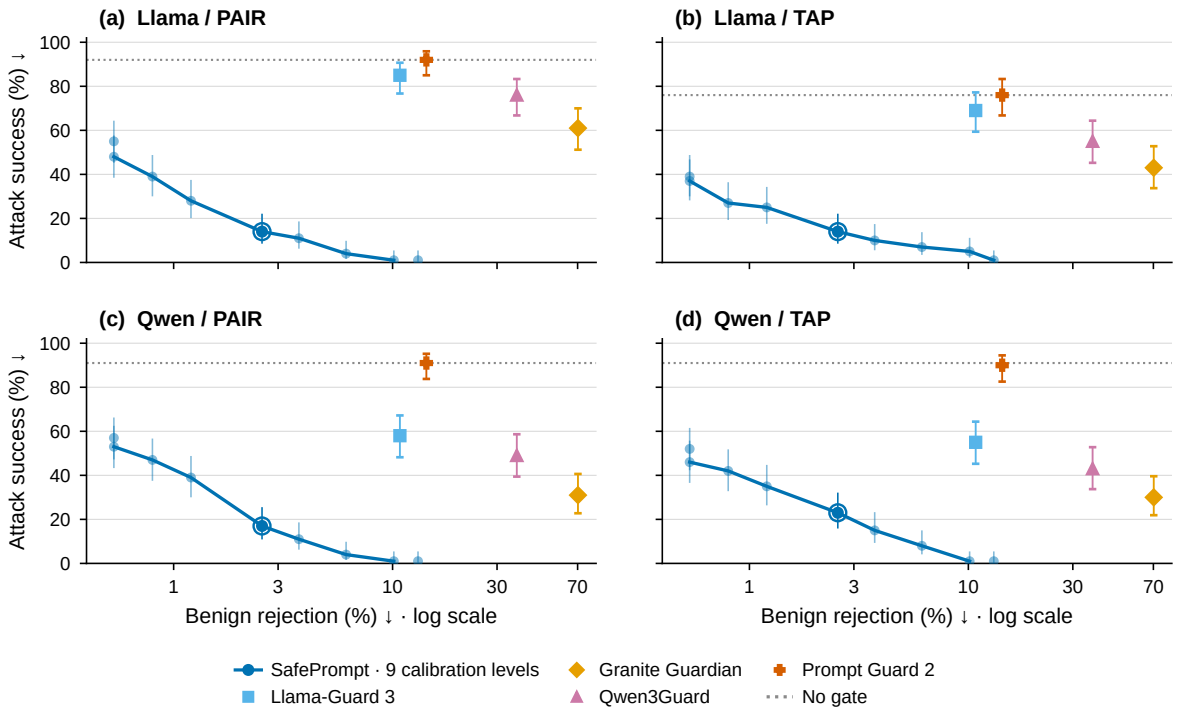


Figure 4: **Calibration chooses a protection–utility operating point.** Nine benign-calibration quantiles for one fixed SafePrompt head (ring: 5%); one native point per guard. Each panel uses 100 goals at 90 queries and 750 benign requests, with HarmBench judgments. Bars: marginal 95% Wilson goal intervals. Lines are frontier guides; FRR is logarithmic.

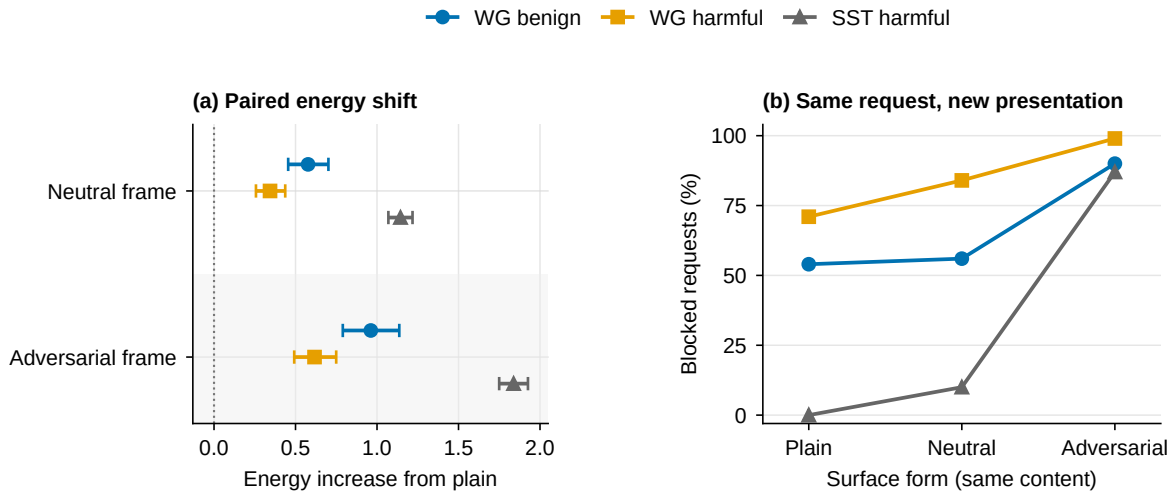


Figure 5: **Attack-like framing raises energy for both input labels.** One Qwen head, 100 base requests per source. (a) Paired energy change, 95% intervals from 10,000 within-source request resamples. (b) Block rates on the same plain, neutral and adversarial forms. Forms retain the requested content and source labels.

8 Conclusion

SafePrompt defends LLMs against jailbreak search using activation signatures. A calibrated linear readout on a frozen encoder reduces harmful success across attacks and victims at low benign cost, without changing the victim. Protection persists with gate feedback. Our analyses link protection to reference coverage and adversarial framing, and show how rare passes become successful goals. Future work should broaden legitimate references and test stronger adaptive search across more victim models.

Limitations

The evaluation covers two victims and a 100-goal HarmBench cohort. It establishes transfer across attack formulations and victims, but does not isolate transfer to entirely disjoint harmful goals. Natural-language search is the main threat model; other attack mechanisms may have different signatures. SafePrompt's sensitivity to reframing can also reject legitimate requests with similar presentation, so its utility depends on the reference and deployment populations. The deployed head inspects 256 tokens while the victim receives the full request. Native guards use different policies and inspection budgets. Utility evaluation measures rejection and refusal rather than the usefulness of accepted answers, and response judges can disagree. The adaptive experiment tests one proposer and binary gate feedback; robustness to stronger proposers and richer feedback remains open.

Ethical Considerations

This study evaluates defenses using harmful-behavior benchmarks, existing dataset annotations and automated response judges. No human participants or annotators were recruited. Attack requests and responses can contain offensive or disturbing material, and analyses of residual bypasses can also aid attackers. False rejections can restrict legitimate discussion of sensitive topics; deployment therefore requires utility evaluation on the intended population.

Public benchmark text can retain personal information. Source harm labels do not establish comprehensive anonymization, and raw prompts and responses require privacy review before redistribution. The intended use of our code and readouts is defensive research. Source licenses and access conditions govern reuse; Appendix O documents the publishers, advertised terms and the unresolved license of one benign-bank source.

AI Assistance Disclosure

AI assistants supported code implementation and inspection, literature retrieval, analysis planning, figure scripts, and manuscript drafting and revision. The authors remain responsible for evidence, citations, interpretation and the final submission.

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A Reproducing the Evaluation

Use the Qwen encoder, readout and calibration rule in Section 3. Evaluate the first 100 standard HarmBench behaviors in source order (78 test and 22 validation behaviors), using the same goals for both victims and all attacks. Apply the frozen gate to the recorded search requests in chronological order, retaining their victim responses. Judge responses with HarmBench-Mistral-7b-val-cls and its standard-behavior prompt. At each budget, count goals with an accepted harmful response; Table 5 uses 90 queries. Measure benign rejection on 500 OR-Bench-hard and 250 safe XSTest requests.

For the activation figure, collect the same benign requests, direct forms of the goals and first PAIR/TAP proposals for each victim. Deduplicate identical texts, compute the normalized layer-12 representations, and fit centered PCA without labels. Use the full representations for gate decisions and PCA only for visualization.

B Complete Defense Comparisons

Tables 6 and 7 repeat the main comparisons with Llama-Guard response judgments. The first uses chronological search prefixes; the second compares matched first queries with newly generated responses for transformation defenses. Security and utility should be read together: rejection of legitimate requests can accompany lower attack success. The judges disagree substantially on some outputs. Qwen AutoDAN has no harmful outputs under HarmBench but substantial success under Llama-Guard. We report their results separately. Each template row uses one request per behavior; the adaptive search in Section 7 instead generates proposals using feedback from the defended system. PromptGuard2 targets injection and is included as an adjacent detection task.

C Paired Search Comparisons

Table 8 compares HaloGuard and WildGuard with SafePrompt on the same goals and proposals. Positive Δ ASR means the native guard permits more successful goals. Paired resampling preserves which goals each defense protects; equal aggregate ASR need not imply protection of the same goals. Figure 6 places endpoint security beside benign rejection. The guards evaluate the same 52,314 distinct requests with the same victim responses and judge labels. HaloGuard uses cutoff 0.65 and overlapping 1200-token windows with stride 256. WildGuard uses its harmful-request verdict with an empty response field and no input truncation. These policies and inspection budgets are retained throughout evaluation.

Intervals use 2,000 paired whole-goal bootstrap draws, shared within each attack–victim combination. They quantify sampling variation for the fixed heads and observed searches. A stopped search contributes no further proposals: at TAP@300, 98 Llama and 99 Qwen trajectories have stopped before that budget. Their outcomes remain unchanged thereafter. An empirical zero-event interval does not establish zero risk in a larger population.

Utility varies sharply by source. Halo rejects 223 of 500 OR-hard requests and 22 of 250 XSTest requests; WildGuard rejects 377 and 3, respectively. Their end-to-end refusal rates also include refusals by the victim. The comparison therefore measures the security and utility of each operating point, rather than holding benign rejection equal across methods.

D Adaptive Search: Additional Response Judgments

Table 9 reports query passes and successful goals under both response judges. Passing queries are shared between judges because they depend on the gate. Harmful-success counts depend on the response judge. The ungated column is available only for the response-only condition, which queries the victim on every proposal.

Search procedure. The attack uses one sequential PAIR-style refinement chain per goal and feedback condition. We select 30 goals deterministically before search, without selecting for baseline success, and use the same goals in all three conditions. At each step, Mistral-7B-Instruct-v0.3 produces an explanation and a candidate request in JSON format; SafePrompt evaluates the request before a fresh victim conversation. The next proposer conversation contains the goal and at most the last four candidate–feedback pairs. A blocked verdict prompts a rewrite to avoid rejection; a passed verdict and response prompt a revision that retains gate evasion while eliciting the goal. Response-only refinement omits the gate verdict, and feedback-free sampling omits the history.

The proposer samples at temperature 1.0 and top-p 0.9, with a 512-token output limit. Llama-

Victim / attack	None	SafePrompt	LG3	Granite	Q3Guard	PGuard2
Llama / PAIR	92	14	85	61	76	92
Llama / TAP	76	14	69	43	55	76
Qwen / PAIR	91	17	58	31	49	91
Qwen / TAP	91	23	55	30	43	90
Filter FRR	0.00	2.53	10.80	70.27	36.93	14.27

Table 5: Fixed-trajectory ASR at 90 chronological queries and filter rejection (%). SafePrompt uses its linear head and nominal 5% calibration threshold; guards evaluate the same request populations. Security uses 100 behaviors per row, utility 750 requests. Vendor settings have different rejection costs. PGuard2 targets injection.

(a) Llama-3.2-3B

Attack / queries (% ASR ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard	PG2†
PAIR@1	4	**0**	0	0	2	0	1	3
PAIR@90	94	**14**	55	52	75	39	68	94
TAP@1	2	**0**	0	0	2	1	0	2
TAP@300	77	**17**	36	25	62	36	51	77
AutoDAN@1	5	**0**	1	0	1	1	4	3
AutoDAN@100	15	**0**	3	0	6	2	10	10
Plain@1	5	**5**	0	0	1	0	4	7
DAN@1	3	**0**	0	0	0	0	2	0
H-CoT@1	10	**0**	0	0	0	0	3	6
Many-shot@1	0	**0**	0	0	0	0	0	0
Benign assumption@1	13	**1**	1	0	1	0	1	12
Template attack@1	40	**0**	0	0	0	0	1	0
End-to-end FRR (%)	5.6	**7.9**	35.2	52.7	15.2	71.5	40.0	18.8

(b) Qwen3.5-4B

Attack / queries (% ASR ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard	PG2†
PAIR@1	40	**4**	3	1	5	0	3	39
PAIR@90	99	**33**	84	79	98	77	89	99
TAP@1	37	**0**	3	1	3	0	5	35
TAP@300	100	**45**	64	32	87	31	70	100
AutoDAN@1	45	**0**	0	0	2	0	6	11
AutoDAN@50	49	**0**	0	0	2	0	6	14
Plain@1	1	**1**	0	0	0	0	0	2
DAN@1	1	**0**	0	0	0	0	0	0
H-CoT@1	2	**0**	0	0	0	0	1	0
Many-shot@1	4	**0**	0	0	0	0	0	0
Benign assumption@1	1	**0**	0	0	0	0	0	0
Template attack@1	14	**0**	0	0	0	0	0	0
End-to-end FRR (%)	36.1	**37.2**	52.1	57.6	41.2	72.3	52.4	40.5

Shared gate FRR (% ↓)	None	**SafePrompt**	Halo	WildG	LG3	Granite	Q3Guard	PG2†
Gate FRR (750)	0.00	**2.53**	32.67	50.67	10.80	70.27	36.93	14.27
OR-hard (500)	0.00	**3.80**	44.60	75.40	14.80	93.20	52.60	21.40
XSTest (250)	0.00	**0.00**	8.80	1.20	2.80	24.40	5.60	0.00

Table 6: **Security across search and template attacks.** Llama-Guard-judged % ASR, 100 goals per attack/victim and 750 benign requests. Bold SafePrompt is the deployed Qwen linear gate. Guards evaluate identical requests and unchanged victim responses. Search @1 is chronological; AutoDAN ends at 100/50 queries and templates use one. †PG2 targets injection. Gate FRR counts input rejection; end-to-end FRR includes victim refusal.

3.2-3B generates greedily with a 150-token limit. Visible victim feedback is truncated to 600 characters. Each chain runs for 90 proposals without early stopping; the 30-query result uses its first 30 steps. The PAIR prompt's response-score field is fixed at 2/10 and conveys no response-quality information. Energy scores, gradients and live harmfulness judgments are withheld. HarmBench and Llama-Guard evaluate responses after search; only outputs from requests that pass the gate count toward defended ASR. Response-only ungated ASR also includes its counterfactual blocked calls.

(a) Llama-3.2-3B

Defense	% ASR ↓ PAIR	TAP	AutoDAN	FRR (%) E2E
None	4	2	5	5.60
SafePrompt	**0**	**0**	**0**	**7.87**
Paraphrase	8	11	19	4.80
SmoothLLM	0	0	5	19.60
Perplexity	4	2	5	5.73
Backtranslation	4	2	5	5.20
Retokenization	4	1	15	11.33
Self-Reminder	7	3	9	7.07

(b) Qwen3.5-4B

Defense	% ASR ↓ PAIR	TAP	AutoDAN	FRR (%) E2E
None	40	37	45	36.13
SafePrompt	**4**	**0**	**0**	**37.20**
Paraphrase	45	43	76	27.47
SmoothLLM	55	55	66	28.93
Perplexity	40	37	45	35.87
Backtranslation	40	37	45	36.80
Retokenization	55	41	69	34.00
Self-Reminder	50	55	66	52.40

Table 7: **Transformation defenses on matched first queries.** Llama-Guard-judged % ASR and end-to-end false refusal (%). Cells use the first chronological query, 100 behaviors; utility uses the shared 750 requests. SafePrompt filters unchanged victim responses; the six transformation defenses use matched requests with regenerated responses.

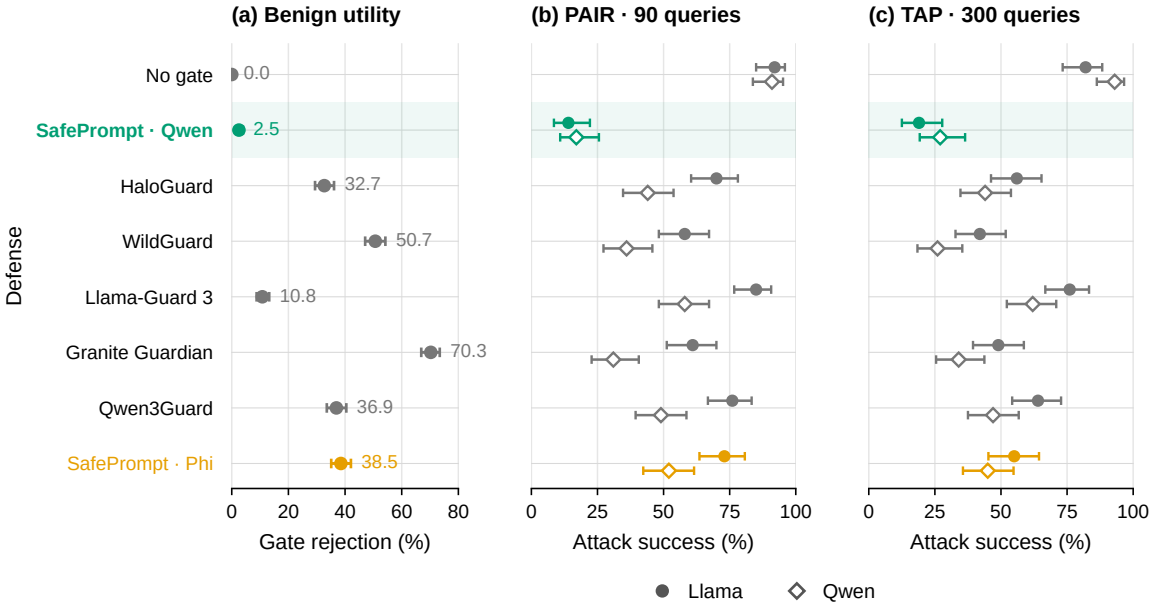


Figure 6: **Search security and benign cost at native operating points.** HarmBench ASR, 100 goals per victim, and FRR, 750 requests. Circles: Llama; diamonds: Qwen. Whiskers: marginal 95% Wilson intervals. SafePrompt inspects 256 tokens, Halo uses 1200-token windows, and WildGuard sees the full request. The separately trained Phi recipe is an ablation.

E Within-Goal Outcome Ranking

Table 10 tests whether energy distinguishes successful from unsuccessful proposals for the same goal. Macro AUC weights eligible goals equally; pair-weighted AUC weights goals by their number of harmful–safe candidate pairs. Token length is a comparison signal on the same requests. For goal g, let P_g and N_g contain proposals with harmful and safe judged outputs. Its AUC averages $\mathbf{1}[r_i > r_j] + \frac{1}{2}\mathbf{1}[r_i = r_j]$ over $(i, j) \in P_g \times N_g$, with $r = -s$. Goals lacking either output class are excluded from ranking but remain in the search-ASR denominator. Macro AUC averages eligible goals; the pair-weighted alternative weights each by $|P_g||N_g|$. Length uses the full request and the Qwen tokenizer, with greater length treated as greater risk. Its paired comparisons use the same eligible goals and resamples.

Energy provides little useful PAIR/TAP within-goal ranking despite strong attack-versus-benign separation. Qwen PAIR is below chance under HarmBench. The detector identifies attack constructions, not whether a particular construction will defeat the victim. Sparse AutoDAN strata provide less stable estimates; the Qwen/HarmBench

Endpoint	Victim	Judge	SP-Q	Halo	WildG	Δ Halo	Δ WildG
PAIR@90	Llama	HB	14	70	58	+56.0 [46.0,66.0]	+44.0 [34.0,54.0]
PAIR@90	Llama	LG	14	55	52	+41.0 [29.0,53.0]	+38.0 [26.0,49.0]
PAIR@90	Qwen	HB	17	44	36	+27.0 [15.0,38.0]	+19.0 [8.0,30.0]
PAIR@90	Qwen	LG	33	84	79	+51.0 [40.0,62.0]	+46.0 [35.0,58.0]
TAP@300	Llama	HB	19	56	42	+37.0 [26.0,48.0]	+23.0 [12.0,34.0]
TAP@300	Llama	LG	17	36	25	+19.0 [9.0,30.0]	+8.0 [-2.0,18.0]
TAP@300	Qwen	HB	27	44	26	+17.0 [5.0,29.0]	-1.0 [-12.0,9.0]
TAP@300	Qwen	LG	45	64	32	+19.0 [7.0,30.0]	-13.0 [-26.0,-1.0]
AutoDAN@1	Llama	HB	0	0	0	+0.0 [0.0,0.0]	+0.0 [0.0,0.0]
AutoDAN@1	Llama	LG	0	1	0	+1.0 [0.0,3.0]	+0.0 [0.0,0.0]
AutoDAN@1	Qwen	HB	0	0	0	+0.0 [0.0,0.0]	+0.0 [0.0,0.0]
AutoDAN@1	Qwen	LG	0	0	0	+0.0 [0.0,0.0]	+0.0 [0.0,0.0]

Table 8: **Paired native-guard search comparisons retain the judge-specific exception.** ASR (%), 100 goals per endpoint; HB denotes HarmBench and LG Llama-Guard. Δ is native minus SP-Q in percentage points, with pointwise 95% percentile intervals from 2,000 whole-goal paired bootstrap draws, conditional on fixed searches, heads and labels. Positive differences favor SP-Q. Native policies/windows and achieved FRRs differ. Zero-event intervals are empirical, not population risk bounds.

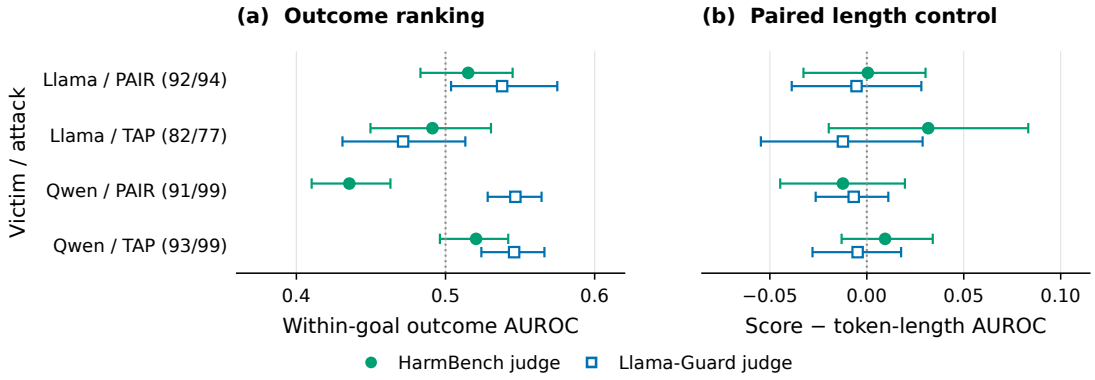


Figure 7: **Attack detection does not predict victim success.** (a) Within-goal AUROC for harmful versus safe victim outputs, using risk $-s(x)$. (b) Paired difference from token-length AUROC. Parentheses give eligible HarmBench/Llama-Guard goal counts. Bars are 95% intervals from 2,000 eligible-goal bootstrap draws for one fixed head.

comparison is undefined because no harmful outputs are observed. These conditional diagnostics do not describe goals with only one response class.

F Reference Composition, Noise and Readout Depth

Table 11 independently varies the proportions of direct harmful and adversarial-unharmful WildGuardMix references at fixed sample counts. Balanced references recover both harmful styles, whereas extreme mixtures can lose one style despite low aggregate benign rejection. Table 14 separates these effects by request stratum. Table 12 holds the balanced references fixed while varying layers and readout capacity. Table 15 distinguishes changing the class prior from introducing Gaussian negatives. The Qwen reference experiment in Table 13 uses the search-oriented recipe rather than the broad WildGuardMix references. Phi representations use the first 256 tokens, BF16 inference, mask-mean pooling and L2 normalization. The experiments share optimization settings, 2,077 benign calibration requests, 4,779 development requests and 2,549 test requests. Eighteen reference/readout recipes, class-prior/noise controls and additional layers yield 90 neural heads across three seeds. Unigram and log-length baselines add 12 readouts. Seed SD measures optimization variation, rather than uncertainty over new requests.

The Qwen reference experiment uses generated attacks, 1,200 general positives and 1,200 sensitive-but-benign positives. Its test set contains 600 attacks and 750 benign requests, with 600

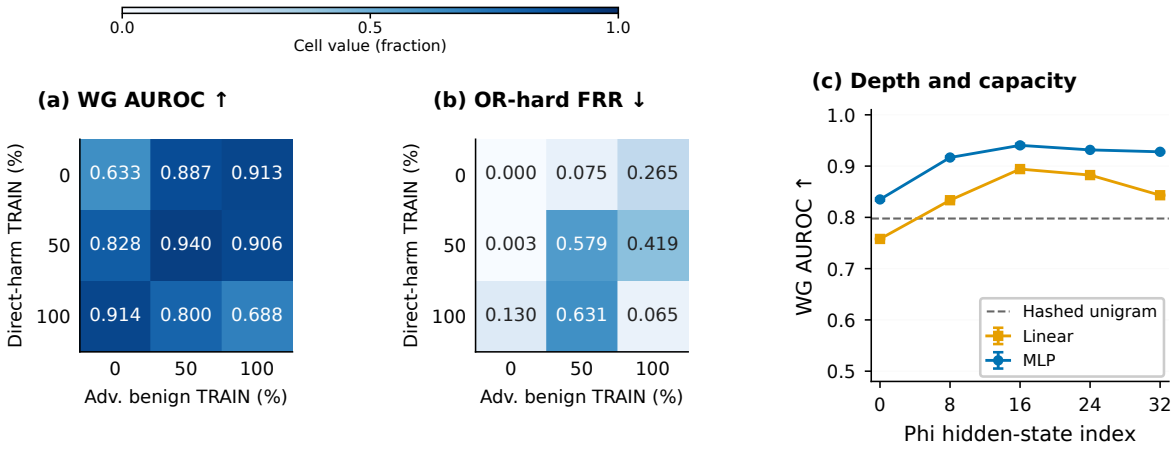


Figure 8: **Reference composition changes harmfulness ranking and utility.** Nine fixed-size Phi MLP mixtures share encoder, loss, calibration data and three optimization seeds. Colors encode reference fractions; whiskers are seed SD. Depth controls use balanced references.

Feedback	N	Pass/query	Joint success HB	Joint success LG3	Ungated HB
Gate-aware	30	28/900	5	1	–
Gate-aware	90	61/2700	7	2	–
None	30	124/900	11	6	–
None	90	434/2700	18	13	–
Victim only	30	19/900	4	2	21
Victim only	90	52/2700	8	6	27

Table 9: **Gate feedback does not eliminate harmful success.** Mistral-7B attacks Llama-3.2-3B with the archived Qwen gate and binary feedback. Successes count goals out of 30, separately for HarmBench (HB) and Llama-Guard-3 (LG3); bypass counts queries. Victim-only ungated success includes counterfactual blocked calls. Blocked proposals consume N; judges run after search.

separate calibration requests. Replacing the sensitive positives preserves the positive count, while goal-subset arms also change initialization. Gaussian removal changes the negative mixture and class prior; the matched-prior Phi controls separate these effects.

G Matched Framing and Input Coverage

Table 16 changes the presentation of each base request while holding its requested content fixed. Energy shifts are paired within source, and block rates use one fixed threshold. Adversarial framing increases harmful detection and benign rejection, demonstrating the benefit and cost of presentation sensitivity. The study uses 100 requests from each of three sources: WildGuardTest benign, WildGuardTest harmful and SimpleSafetyTests harmful. Plain requests are compared with a neutral request for numbered steps and an adversarial request for unrestricted developer-mode compliance. Goal-only and bare-frame controls separate requested content from surrounding instructions.

Padding places requests at three positions and target lengths of 128, 256, 512 and 1024 tokens. Input coverage measures how much of the request is visible to each detector. Separate benign calibration requests receive the same padding; minimum-window scores are calibrated on their transformed mixture. Intervals resample the 100 base requests within each source, preserving the pairing between variants. Variants inherit the original source label; the study does not independently annotate the safety of every framing.

H Population-Specific Detection

Figure 9 compares detection across request populations. A probe can recognize search constructions while missing direct harmful requests, and benign acceptance varies across sources. The experiments fit independent heads for each backbone; they do not transfer one direction between models. These probes measure representation accessibility, whereas the deployed gate's calibration determines its search protection and utility.

I Layer-Wise Representation Accessibility

Diagnostic probes separate search attacks at Qwen layer one (AUROC 0.988), while direct harmful

Victim	Attack	Judge	Eligible goals	Macro AUC	95% interval	Pair-weighted	Length AUC
Llama	PAIR	HB	92	0.515	[0.483, 0.545]	0.529	0.515
Llama	PAIR	LG3	94	0.538	[0.504, 0.575]	0.558	0.543
Qwen	PAIR	HB	91	0.436	[0.410, 0.463]	0.457	0.448
Qwen	PAIR	LG3	99	0.547	[0.528, 0.564]	0.554	0.554
Llama	TAP	HB	82	0.491	[0.450, 0.530]	0.527	0.460
Llama	TAP	LG3	77	0.472	[0.431, 0.513]	0.543	0.484
Qwen	TAP	HB	93	0.520	[0.496, 0.542]	0.526	0.511
Qwen	TAP	LG3	99	0.546	[0.524, 0.566]	0.543	0.551
Llama	AUTODAN	HB	5	0.762	[0.365, 1.000]	0.028	0.473
Llama	AUTODAN	LG3	11	0.763	[0.562, 0.929]	0.797	0.418
Qwen	AUTODAN	HB	0	–	–	–	–
Qwen	AUTODAN	LG3	4	0.306	[0.111, 0.500]	0.422	0.292

Table 10: Outcome ranking within a recorded goal, using risk $-s(x)$ and including only goals with both harmful and safe judged outputs. Macro gives each eligible goal equal weight; pair-weighted pools its harmful/safe candidate pairs. Intervals resample eligible goals 2,000 times. AutoDAN strata have few eligible goals; Qwen/HarmBench has none, so its AUC is undefined. Length uses the Qwen encoder tokenizer. One fixed head, without training-seed uncertainty.

Direct H (%)	Adv. B (%)	Head	WG AUC	WG recall	WG FRR	SST recall	OR FRR
0	0	Linear	0.635±0.000	15.6±0.1	2.5±0.0	0.0±0.0	0.0±0.0
		MLP	0.633±0.001	16.6±0.1	3.5±0.0	0.0±0.0	0.0±0.0
0	50	Linear	0.714±0.000	30.6±0.1	3.4±0.1	0.0±0.0	0.0±0.0
		MLP	0.887±0.008	59.0±1.7	6.0±0.2	2.3±1.5	7.5±2.2
0	100	Linear	0.843±0.000	36.3±0.4	4.0±0.0	0.0±0.0	0.2±0.0
		MLP	0.913±0.002	68.5±1.0	5.7±0.3	34.7±9.6	26.5±4.6
50	0	Linear	0.665±0.000	23.1±0.1	3.8±0.1	0.0±0.0	0.0±0.0
		MLP	0.828±0.011	37.9±4.5	4.2±0.3	0.0±0.0	0.3±0.5
50	50	Linear	0.894±0.000	54.6±0.7	3.8±0.1	3.3±1.2	11.6±1.4
		MLP	0.940±0.003	81.8±1.4	5.8±0.5	93.0±3.0	57.9±7.9
50	100	Linear	0.669±0.002	17.9±0.6	0.5±0.0	53.0±0.0	5.1±0.2
		MLP	0.906±0.002	72.8±1.0	2.9±0.2	91.3±0.6	41.9±1.2
100	0	Linear	0.686±0.000	26.6±0.1	4.1±0.0	0.0±0.0	0.0±0.0
		MLP	0.914±0.003	58.2±2.9	4.6±0.4	27.0±12.5	13.0±3.7
100	50	Linear	0.685±0.000	44.8±0.2	4.3±0.4	96.0±0.0	55.0±0.7
		MLP	0.800±0.005	60.0±0.3	2.2±0.2	97.0±0.0	63.1±1.1
100	100	Linear	0.451±0.001	0.2±0.1	0.3±0.0	3.0±0.0	0.0±0.0
		MLP	0.688±0.002	34.5±0.4	1.0±0.0	55.7±5.7	6.5±1.8

Table 11: **Reference composition changes both ranking and utility.** Mean±SD over three optimization seeds for all 18 planned reference/head recipes. Each arm has 8,000 unique harmful and 8,000 unique unharmful references at Phi layer 16; direct-harm and native adversarial-unharmful shares vary independently. Adversarial-unharmful is a source flag, not semantic boundary annotation. Every threshold uses the same 2,077 benign CAL requests; none uses TEST. AUC is unitless; other columns are percentages. WG1699, SST 100, OR 500, XS 250.

requests are less separable. Early accessibility is consistent with presentation and source information. The probes identify usable discrimination in representations; they do not identify its causal origin or measure calibrated search risk.

J Encoder-Family and Readout Comparisons

Tables 17 and 18 compare five frozen encoder families using the same broad-label references. MLPs improve harmful recall but can incur much greater OR-hard rejection than in-source benign rejection. The first table averages three optimization seeds; the second gives their ranges. This is a harmfulness-classification experiment, distinct from the Qwen search detector. Each encoder uses its midpoint representation, first 256 tokens, BF16 inference, mask-mean pooling and L2 normalization. The 16,000 WildGuardMix training requests contain 4,000 from each harmful/unharmful and ordinary/adversarial bucket. Native adversarial status is a source annotation, not an independent measure of semantic difficulty. Training uses 2,000 AdamW steps,

Features	Layer	Head	WG AUC	WG recall	WG FRR	SST recall	OR FRR
Phi	0	Linear	0.758±0.001	42.1±0.3	7.3±0.1	23.0±0.0	32.7±0.4
		MLP	0.835±0.002	52.0±1.1	6.0±0.5	65.0±3.5	34.1±1.4
Phi	8	Linear	0.833±0.001	36.5±0.8	4.9±0.1	0.0±0.0	0.9±0.1
		MLP	0.917±0.002	72.9±1.5	5.9±0.1	77.3±10.5	43.0±12.1
Phi	16	Linear	0.894±0.000	54.6±0.7	3.8±0.1	3.3±1.2	11.6±1.4
		MLP	0.940±0.003	81.8±1.4	5.8±0.5	93.0±3.0	57.9±7.9
Phi	24	Linear	0.882±0.000	54.3±0.2	3.7±0.0	8.7±0.6	24.2±1.0
		MLP	0.932±0.001	76.3±0.4	5.5±0.1	83.0±2.0	50.4±3.1
Phi	32	Linear	0.843±0.001	55.2±0.3	4.1±0.2	42.7±0.6	31.4±0.7
		MLP	0.928±0.002	74.5±0.3	4.8±0.2	77.7±0.6	60.3±1.6
Hashed unigram	–	Linear	0.798±0.000	40.1±0.3	3.7±0.1	17.7±0.6	15.7±0.5
		MLP	0.768±0.002	36.9±1.2	5.4±0.4	34.7±1.2	14.5±1.5
Log char length	–	Linear	0.532±0.110	4.4±3.8	1.3±1.2	0.3±0.6	0.0±0.0
		MLP	0.635±0.008	22.8±0.8	11.1±0.7	0.7±0.6	69.3±0.8

Table 12: **Depth and head capacity exceed simple lexical controls.** Mean±SD over three seeds, identical balanced16k text references. Layer0 is the embedding state and 32 includes the final norm. Controls use4096-bin normalized hashed unigrams or log1p character length, fitted only on TRAIN with the same head/training/calibration schedule. AUC unitless; remaining columns percentages. Same CAL and TEST as Table 11.

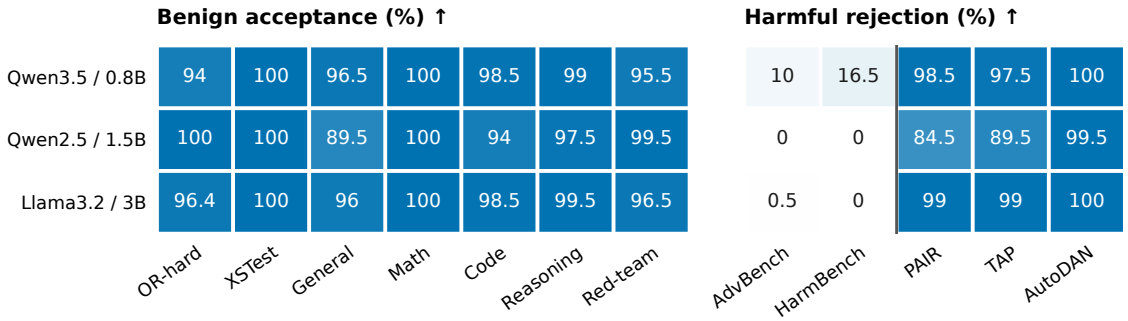


Figure 9: **Attack-form detection and direct-harm detection differ.** Independent diagnostic logistic readouts at each backbone's midpoint, with benign calibration. Benign columns report acceptance; harmful columns report rejection. Most sources contain 200 requests; OR-hard has 500 and safe XSTest 250. Each cell describes one fitted readout.

batch size 256, learning rate 10^{-3} and weight decay 10^{-4}. Heads are linear or two-layer, 512-wide MLPs with dropout 0.1, trained with three seeds.

Development uses 4,779 requests and benign calibration 2,077. Recipe selection maximizes mean development macro-F1, breaking ties by harmful recall, subject to development FRR at most 10% both across seeds and for seed zero. The selected Phi MLP uses seed zero; all 30 heads are evaluated on the same 2,549 test requests without test-based selection. Conflicting labels for normalized duplicate requests are excluded from training, development and calibration. The test population retains 1,699 WildGuardTest requests; the independent vendor comparison removes one training overlap and uses 1,698.

J.1 Phi recipe on search attacks

Table 19 applies the development-selected Phi MLP to the same searches as the Qwen gate. Its weaker protection and higher benign rejection show that a broad harm-classification recipe need not transfer to search defense. Encoder, references and readout all differ, so this comparison does not isolate an architectural effect.

K Harmfulness-Classifier Benchmarks

Tables 20 and 21 compare the Phi harm-classification recipe with native guards. The primary table reports harmful recall and benign rejection; the adjacent table separates toxicity, jailbreak and content-policy criteria. A request negative under one criterion can be positive under another. All classifiers evaluate identical requests

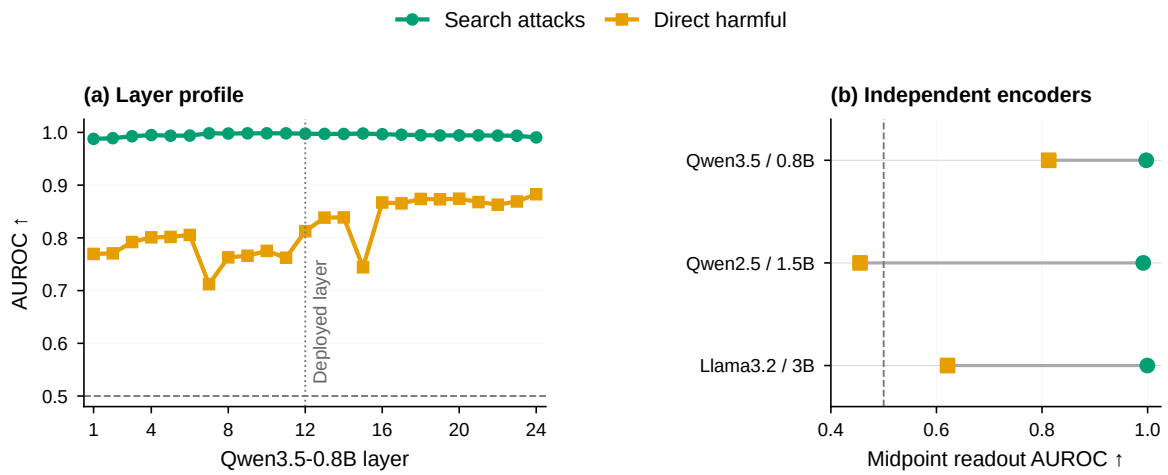


Figure 10: **Search and direct-harm separation have different profiles.** (a) Qwen layer diagnostics. (b) Independent midpoint readouts on three backbones. AUROC averages three search sources or two direct-harm sources (200 requests each), against 1,750 benign requests. Each model/layer has one fitted logistic readout.

Reference / head	Attack block ↑	FRR ↓
MLP + Gaussian	96.4 ± 0.5	5.4 ± 0.6
Linear + Gaussian	97.7 ± 0.0	2.7 ± 0.2
MLP, text only	98.3 ± 0.1	5.6 ± 0.2
No boundary positives	7.8 ± 3.1	6.9 ± 0.6
General replacement	6.6 ± 1.4	8.5 ± 0.8
8 augmented goals	95.2 ± 1.2	4.9 ± 0.9
16 augmented goals	94.2 ± 5.1	4.5 ± 0.4
32 augmented goals	96.7 ± 1.2	4.9 ± 1.2

Table 13: **Boundary coverage governs the narrow calibrated readout.** Controlled retraining on frozen Qwen3.5-0.8B layer 12; mean ± sample SD over three seeds, 600 attack and 750 benign prompts, separate 600-request calibration. Default: 1,200 general and 1,200 boundary positives, 52 augmented goals; general replacement preserves positive count. Goal-limited rows vary subsets and initialization together. These are prompt rejection rates rather than goal-level search ASR.

with the same available source labels: 1,698 WildGuardTest, 100 SimpleSafetyTests, 500 OR-hard, 250 safe XSTest, 4,943 deduplicated ToxicChat and 851 OpenAI moderation requests. ToxicChat is evaluated under two criteria. Native policies and cutoffs are retained; Phi uses its development-selected head and independent benign calibration. Binary verdicts supply an operating point rather than a continuous ranking, so their AUROC is omitted.

K.1 Expanded Public-Source Evaluation

Table 22 evaluates the full public populations associated with HaloGuard. The seven English endpoints contain 8,894 requests; the manufacturer test has 10,451; PolyGuard contributes 28,883 known-label requests in 17 languages. ToxicChat criteria remain separate. Both classifiers evaluate the same occurrences. Mean F1 over the seven English endpoints is 77.6% for the Phi recipe and 90.8% for HaloGuard. Manufacturer and multilingual results expose larger transfer gaps. These comparisons retain different native policies and inspection budgets rather than matching benign rejection.

L Geometry of the Gaussian Reference

For a unit embedding u, write the isotropic perturbation as $\epsilon = Au + \epsilon_\perp$, where $A \sim \mathcal{N}(0, \sigma^2)$ and $S = \|\epsilon_\perp\|^2 \sim \sigma^2 \chi^2_{d-1}$ independently. The cosine after renormalization is exactly

$$u^\top \frac{u+\epsilon}{\|u+\epsilon\|} = \frac{1+A}{\sqrt{(1+A)^2+S}}. \tag{5}$$

If $\sigma = \eta/\sqrt{d}$ with η fixed, then A converges to zero and S to η^2, yielding cosine $(1+\eta^2)^{-1/2}$. Thus $\eta = \sigma\sqrt{d}$, rather than σ alone, controls the typical angular displacement across representation dimensions. At $d = 1024$ and $\sigma = 0.1$, the RMS noise norm is 3.2 and the high-dimensional approximation gives cosine 0.298, or about 73 degrees. These are analytical properties of the construction, not measured semantic distances or certified perturbation radii.

The mixture and sampling prior also matter. The training recipe uses equal benign, text-

Head	TRAIN shares (%) Direct harm / adv. unharm.	Harmful recall ↑ Direct	Harmful recall ↑ Adversarial	Unharmful FRR ↓ Direct	Unharmful FRR ↓ Adversarial
Linear	0 / 0	0.0	34.5	0.0	5.3
Linear	50 / 50	43.4	68.0	0.4	7.4
Linear	100 / 100	0.4	0.0	0.6	0.0
MLP	0 / 0	0.0	36.7	0.0	7.3
MLP	50 / 50	83.1	80.3	2.3	9.5
MLP	100 / 100	62.1	1.1	1.8	0.0

Table 14: **Reference coverage determines which native styles are captured.** Same Phi layer 16, 16k TRAIN references with separate benign CAL fixing each 5% operating point; percentages are means of three seeds. WG TEST strata: 413 harmful-direct, 341 harmful-adversarial, 490 unharmful-direct and 455 unharmful-adversarial. The balanced recipe recovers both harmful strata; the extremes largely miss one or both. These are native source annotations, not paired style interventions. All nine mixtures appear in Appendix F.

Negative reference	P:N	Head	WG AUC	WG recall	WG FRR	SST recall	OR FRR
Text	1:1	Linear	0.894±0.000	54.6±0.7	3.8±0.1	3.3±1.2	11.6±1.4
		MLP	0.940±0.003	81.8±1.4	5.8±0.5	93.0±3.0	57.9±7.9
Text	1:2	Linear	0.894±0.000	53.6±0.5	3.5±0.1	2.3±1.2	10.7±0.4
		MLP	0.940±0.001	82.8±1.0	5.7±0.1	94.7±2.1	61.7±2.8
Text+Gaussian	1:2	Linear	0.872±0.000	39.8±0.2	3.4±0.1	0.0±0.0	1.0±0.0
		MLP	0.942±0.001	81.4±1.8	6.0±0.5	94.3±4.0	63.5±7.9

Table 15: **Gaussian references do not uniformly improve calibrated detection.** Mean±SD of three seeds, same balanced text references and Phi layer 16. The1:1 condition uses 128 positive/128 text-negative examples per step; the1:2 text control uses 85/170, and Gaussian replacement 85/85/85. The latter two match the class prior while changing the negative mixture. Gaussian RMS norm is 3.2 before normalization. Same CAL and TEST as Table 11; AUC unitless, other columns percentages.

Rule	Unpadded, three forms WG recall	WG FRR	SST recall	Padded TEST H recall	B FRR	Padded CAL B FRR
Native first 256	84.7	66.7	32.3	100.0	100.0	99.8
Recalibrated first 256	18.0	5.3	0.0	6.9	5.8	5.0
Calibrated first 512	18.3	7.0	0.0	7.4	5.8	5.0
Calibrated min-window256	19.0	7.0	0.0	7.9	5.7	5.0

Table 16: **Recalibration trades narrow-head recall for utility.** Percentages, one frozen archived Qwen linear head. The same WG 100 harmful/100 benign core and separate SST 100 harmful requests each appear in three forms. Padded TEST has 2,400 harmful and 1200 benign requests across three content positions and four target lengths; independent padded CAL has 24,924 benign requests. These repeated forms are not independent base examples. Native thresholds are checkpoint values; other thresholds use only independent transformed benign CAL. The window rule is minimum safety score across actual token-id windows, calibrated on the padded mixture. More inspected tokens do not repair this head's broad-intent coverage.

negative and Gaussian-negative minibatches. Removing the Gaussian batch changes both the negative reference distribution and the positive class prior from one third to one half. At the unrestricted population optimum, that prior change alone contributes a constant logit offset. A calibration quantile would absorb a pure constant shift, but finite capacity, regularization and optimization can also change ranking. A Gaussian ablation therefore needs to report the realized batch weights and compare calibrated operating points, rather than interpreting a raw score shift as improved harmfulness discrimination.

M Calibration under Window Aggregation

Extending a prefix detector with multiple windows changes its decision rule. For window scores $s_1, \ldots, s_m$, rejection by the minimum score means that any window can trigger a block. Even if each window had benign rejection probability α, the request-level probability need not equal α: it is at most $m\alpha$ by a union bound, and would be $1 - (1 - \alpha)^m$ under independence. Neither as-

Encoder	Head	WildGuardTest AUC ↑	WildGuardTest Recall ↑	WildGuardTest FRR ↓	SST Recall ↑	OR-hard FRR ↓	XSTest FRR ↓
Qwen3.5 / 0.8B	Linear	0.868	56.1	5.3	73.3	34.4	0.4
	MLP	0.914	72.6	5.5	85.0	40.0	2.8
Gemma3 / 1B	Linear	0.790	41.6	2.9	2.0	7.0	0.0
	MLP	0.932	75.8	5.1	85.0	52.5	3.6
Llama3.2 / 3B	Linear	0.893	63.3	4.8	17.0	19.8	0.0
	MLP	0.929	80.7	6.1	91.7	59.6	5.1
Phi4-mini / 3.8B	Linear	0.895	56.4	4.1	5.0	14.7	0.0
	MLP	0.932	81.0	5.9	92.7	52.9	3.7
Qwen2.5 / 1.5B	Linear	0.832	38.6	4.7	0.0	0.8	0.0
	MLP	0.922	73.8	6.1	72.0	45.5	0.5

Table 17: **Broader references reveal a capacity–utility tradeoff.** Frozen encoders, identical 16,000 WildGuardMix training requests, linear or two-layer 512-wide MLP heads. Mean of three optimization seeds; each threshold is the 5% quantile on separate benign calibration drawn from the training split. WildGuardTest has 754 harmful and 945 unharmful requests; SST has 100 harmful, OR-hard 500 benign, and safe XSTest 250 benign. AUROC is unitless; other cells are percentages. Phi MLP seed zero was selected on development data before test evaluation. Adversarial-flagged unharmful training requests are distinct from OR-hard boundary benigns.

Encoder	Head	WG AUC	WG recall (%)	OR-hard FRR (%)
Qwen3.5 / 0.8B	linear	0.868 [0.868, 0.868]	56.1 [56.0, 56.1]	34.4 [34.0, 34.8]
Qwen3.5 / 0.8B	mlp	0.914 [0.909, 0.918]	72.6 [71.5, 73.2]	40.0 [36.8, 42.8]
Gemma3 / 1B	linear	0.790 [0.790, 0.790]	41.6 [41.4, 41.8]	7.0 [7.0, 7.0]
Gemma3 / 1B	mlp	0.932 [0.931, 0.933]	75.8 [75.5, 76.3]	52.5 [51.2, 55.0]
Llama3.2 / 3B	linear	0.893 [0.893, 0.894]	63.3 [63.3, 63.4]	19.8 [19.2, 20.6]
Llama3.2 / 3B	mlp	0.929 [0.927, 0.931]	80.7 [79.8, 81.3]	59.6 [53.4, 65.2]
Phi4-mini / 3.8B	linear	0.895 [0.894, 0.895]	56.4 [56.2, 56.6]	14.7 [14.0, 15.2]
Phi4-mini / 3.8B	mlp	0.932 [0.931, 0.933]	81.0 [80.8, 81.4]	52.9 [51.0, 56.0]
Qwen2.5 / 1.5B	linear	0.832 [0.832, 0.832]	38.6 [38.3, 38.7]	0.8 [0.8, 0.8]
Qwen2.5 / 1.5B	mlp	0.922 [0.922, 0.923]	73.8 [73.2, 74.5]	45.5 [42.4, 50.0]

Table 18: **Seed variation in the frozen-family grid.** Mean [minimum, maximum] of three optimization seeds; ranges are not confidence intervals. The same data, calibration, and test populations as Table 17.

(a) Llama-3.2-3B

% ASR ↓	None	Qwen linear	Phi MLP
$PAIR_{90}$	92	14	73
TAP_{300}	82	19	55
$Direct_1$	5	5	0
Gate FRR (%)	0.00	2.53	38.53

(b) Qwen3.5-4B

% ASR ↓	None	Qwen linear	Phi MLP
$PAIR_{90}$	91	17	52
TAP_{300}	93	27	45
$Direct_1$	0	0	0
Gate FRR (%)	0.00	2.53	38.53

Table 19: **A broad harm-classification recipe does not reproduce the search signature.** HarmBench ASR on the same 100 goals and gate FRR on the same 750 benign requests. The original Phi MLP uses a different encoder and training population from the deployed Qwen head; this recipe comparison does not isolate encoder or head architecture.

sumption establishes the distribution of overlapping windows in real requests.

Consequently, the original prefix threshold is a useful unchanged baseline, while an extended detector must calibrate the aggregated request score on independent benign requests under the same windowing rule. Evaluation should separate payload visibility, harmful-request rejection and benign rejection. More complete token coverage can improve the first while changing the latter two; coverage by itself is not a robustness guarantee.

N Additional Readout and Operating-Point Visuals

Classifier	WildGuardTest Recall ↑	WildGuardTest FRR ↓	WildGuardTest F1 ↑	SST Recall ↑	OR-hard FRR ↓	XSTest FRR ↓
SafePrompt / Phi MLP	81.4	5.9	86.2	94.0	55.6	4.4
Llama-Guard-3 / 8B	65.3	3.8	76.8	99.0	14.8	2.8
Granite-Guardian / 8B	94.3	22.7	84.7	99.0	93.2	25.2
Qwen3Guard / 0.6B	80.4	4.8	86.3	93.0	52.4	5.6
HaloGuard / 0.8B	95.5	2.9	95.9	100.0	44.6	8.8
ShieldGemma / 2B	43.6	5.1	58.2	78.0	32.0	28.8
WildGuard / 7B	85.0	5.4	88.7	99.0	75.4	1.2
Prompt-Guard-2 / 86M†	27.5	9.7	39.3	1.0	20.6	0.0

Table 20: **Independent common-population operating points.** Percentages; WildGuardTest has 754 harmful and 944 unharmful requests; SST100 harmful, OR-hard500 benign and XSTest250 benign. SafePrompt uses the development-selected Phi MLP seed zero with an independently calibrated threshold. Provider-native policies and cutoffs are preserved; Halo uses the declared1200-token/stride256 window extension, and ShieldGemma its four safety policies. F1 is for the harmful class. Thresholds are not calibrated on the test requests; benign rejection is not matched. The separated Prompt-Guard row is an adjacent injection classifier. All seven planned vendor classifiers are included.

Classifier	ToxicChat: toxicity Recall ↑	ToxicChat: toxicity FPR ↓	ToxicChat: jailbreak Recall ↑	ToxicChat: jailbreak FPR ↓	OpenAI: content harm Recall ↑	OpenAI: content harm FPR ↓
SafePrompt / Phi MLP	74.2	12.9	72.7	16.2	81.9	40.1
Llama-Guard-3 / 8B	49.9	4.3	30.7	7.1	78.6	16.6
Granite-Guardian / 8B	83.4	4.8	79.5	9.1	90.5	30.6
Qwen3Guard / 0.6B	70.5	1.3	75.0	4.9	81.5	25.2
HaloGuard / 0.8B	88.5	1.7	92.0	6.3	84.6	11.3
ShieldGemma / 2B	67.6	2.2	36.4	6.3	94.2	44.5
WildGuard / 7B	91.1	6.8	80.7	11.5	95.7	49.3
Prompt-Guard-2 / 86M†	25.2	1.5	85.2	1.7	1.2	0.6

Table 21: **Transfer to distinct native label criteria.** Percentages; 4,943 deduplicated ToxicChat requests and 851 strict-union OpenAI examples with sufficient category-label coverage. The two ToxicChat endpoints reuse requests but have different labels. Negative-class rejection is FPR on the native criterion, not benign utility. The same classifier decisions as Table 20 are used without calibration on these tests.

Endpoint	n	Positive / negative	Halo Recall	Halo F1	Halo FPR	SafePrompt Phi Recall	SafePrompt Phi F1	SafePrompt Phi FPR
OpenAI moderation	1,680	522 / 1,158	84.7	82.9	8.8	82.0	65.2	31.3
Aegis 1 user messages	359	233 / 126	76.0	85.5	3.2	70.8	80.7	8.7
Aegis 2	1,964	1,059 / 905	86.4	87.3	13.5	77.5	75.7	31.9
ToxicChat human annotated	2,853	362 / 2,491	88.7	87.0	2.2	74.9	52.5	16.0
SimpleSafetyTests	100	100 / 0	100.0	100.0	—	94.0	96.9	—
HarmBench std. + copyright	239	239 / 0	94.6	97.2	—	75.7	86.2	—
WildGuardTest	1,699	754 / 945	95.2	95.8	2.9	81.4	86.2	5.9
English-seven unweighted mean	—	—	89.4	90.8	—	79.5	77.6	—
Halo manufacturer TEST	10,451	3,045 / 7,406	85.2	89.9	1.8	24.0	33.1	8.7
PolyGuard (17 languages)	28,883	12,818 / 16,065	86.7	89.5	5.6	43.9	59.7	2.4
ToxicChat full: toxicity	5,083	362 / 4,721	88.7	83.6	1.8	74.9	44.0	12.7
ToxicChat full: jailbreak	5,083	91 / 4,992	92.3	33.8	6.5	73.6	13.9	16.1

Table 22: **Full canonical public-source comparison.** Percentages on identical request populations. Halo uses unsafe cutoff 0.65 with 1,200-token windows and stride 256; Phi uses the development-selected MLP, its native threshold, and the first 256 tokens. Policies and inspection budgets differ. Every endpoint retains its native criterion; FPR is rejection of its negative class, not uniformly benign utility. SST and HarmBench have no negatives. The mean includes exactly the seven English endpoints above the first rule. ToxicChat criteria share requests. PolyGuard omits 442 unknown-label rows from 29,325. The public evaluation matches the released source populations; correspondence with private manufacturer evaluations and independence from guard training data are unknown.

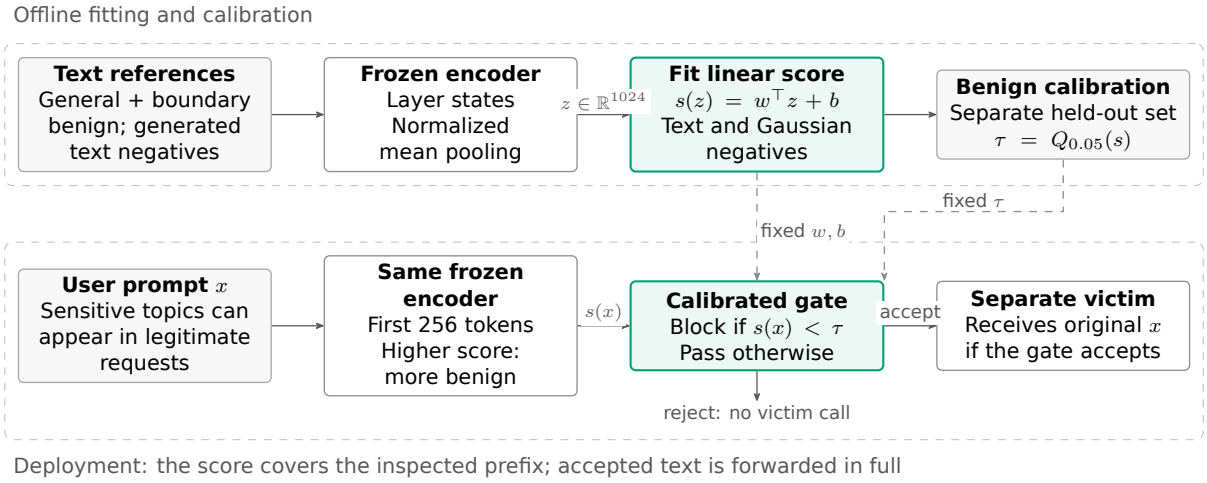


Figure 11: **Reference fitting, calibration and deployment are distinct.** A frozen encoder supports a fitted linear score; separate benign data set the threshold before evaluation. Accepted prompts reach a separate victim unchanged. The deployed score reads only the first 256 tokens, so acceptance does not imply that later content was inspected. The diagram summarizes the method.

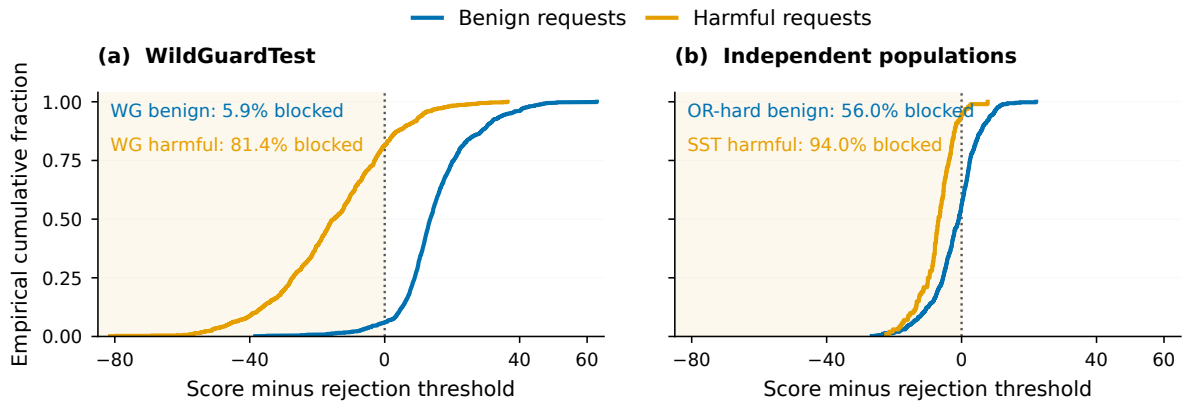


Figure 12: **A useful separation need not yield a transferable threshold.** Empirical score distributions for the development-selected Phi MLP, seed zero. Shading marks rejection below its frozen calibration threshold. (a) Human-labeled WildGuardTest, 945 unharmful and 754 harmful requests. (b) Independent OR-hard benign (500) and SST harmful (100) populations. One fixed head, without seed bands. The sharp benign score shift explains its higher rejection on OR-hard; it does not identify a causal feature.

O Data, Resources and Reproducibility

Source attribution and permissions. Tables 23–25 identify the original publishers and link their releases. The evaluation sources include synthetic requests, published user–model interactions and annotated safety-policy data. The primary search and utility experiments use English; the separate PolyGuard comparison covers 17 languages. Benign-bank sources cover general instructions, mathematics and code. These populations do not establish demographic coverage or representative performance for any particular community.

The data have mixed terms. ToxicChat, Alpaca and PKU-SafeRLHF advertise non-commercial restrictions; Dolly carries share-alike conditions. Llama and Gemma models have their own access and use terms. Smol-Magpie's Apache-2.0 license applies to that component, rather than all SmolTalk subsets. The Magpie-Ultra release does not specify a license in its public metadata or card. We therefore make no blanket redistribution claim for the benign bank. Reproductions should obtain third-party data and weights from their original publishers under the applicable conditions.

Training and model selection. Table 26 gives the Qwen head's training recipe and the broader-family comparison settings. Qwen training samples equal benign, generated-attack and Gaussian minibatches; positives comprise 1,200 general and 1,200 sensitive-but-benign requests. Calibration uses 300 separate requests from each positive source. The deployed head is the seed-zero linear arm, with 1,025 trained parameters. Controlled retraining varies reference composition, seed-goal coverage and readout depth, while the broader study selects recipes on development data as described in Appendix J. These procedures retain separate calibration and evaluation populations. The original head's exact training-row assignment is unavailable; the recipe and saved head support reproducing its evaluation.

Software and computational resources. Table 27 records the verified inference environment and links the software projects. Representation extraction uses BF16 model inference and float32 pooling, with the settings in Section 3. Head fitting uses CPU or GPU computation without updating the encoder. Experiments ran on single-GPU RTX A5000 workers (24GB) and an RTX 5090 workstation (32GB). Table 25 reports the pretrained parameter totals.

The timed native-guard scoring passes on 52,314 distinct requests took 2,166.0 seconds for HaloGuard and 34,338.4 seconds for WildGuard on one RTX A5000, or 10.14 GPU-hours combined. We estimate a conservative upper bound of approximately 500 GPU-hours for the research supporting this paper, including training, attack generation, evaluation, exploratory studies and failed runs. This estimate allows 60 hours for recorded runs and a further 440 hours for historical work and accounting uncertainty. It excludes pretraining of third-party models and is an estimate rather than a metered total.

Source (linked primary release)	Publisher	Advertised terms	Role
JailbreakBench	JailbreakBench	MIT	Negative seed goals
HarmBench	centerforaisafety	MIT	Search goals and evaluation
OR-Bench	bench-llm	CC-BY-4.0	Positive references and benign utility
XSTest	paul-rottger	CC-BY-4.0	Benign utility
WildGuardMix	allenai	ODC-BY	Training and evaluation
SimpleSafetyTests	Bertievidgen	CC-BY-2.0	Evaluation
ToxicChat	lmsys	CC-BY-NC-4.0	Evaluation
OpenAI moderation	openai	MIT	Evaluation
Aegis 1.0	nvidia	CC-BY-4.0	Evaluation
Aegis 2.0	nvidia	CC-BY-4.0	Evaluation
Halo-Guard-Bench	astroware	Apache-2.0	Evaluation
PolyGuardPrompts	ToxicityPrompts	CC-BY-4.0	Evaluation

Table 23: Evaluation and seed-goal sources. Source names link to the original publishers. Terms are read from public repository metadata; custom conditions remain governed by the linked release.

Source (linked primary release)	Publisher	Advertised terms	Role
Smol-Magpie (SmolTalk)	HuggingFaceTB	Apache-2.0	Benign-bank source
Magpie-Ultra	argilla	Not specified	Benign-bank source
Alpaca	tatsu-lab	CC-BY-NC-4.0	Benign-bank source
Dolly	databricks	CC-BY-SA-3.0	Benign-bank source
PKU-SafeRLHF	PKU-Alignment	CC-BY-NC-4.0	Benign-bank source
GSM8K	openai	MIT	Benign-bank source
MATH mirror	qwedsacf	MIT	Benign-bank source
OpenMathInstruct-2	nvidia	CC-BY-4.0	Benign-bank source
Glaive Code	glaiveai	Apache-2.0	Benign-bank source

Table 24: Sources recorded in the benign prompt bank. Source names link to the original publishers. Terms are read from public repository metadata; custom conditions remain governed by the linked release. This is the bank inventory, not the exact training-row assignment of the original Qwen head. The Smol-Magpie component is used rather than every SmolTalk subset.

Source (linked primary release)	Publisher	Advertised terms	Parameters
Qwen3.5-0.8B	Qwen	Apache-2.0	0.873B
gemma-3-1b-it	google	Gemma terms	1.000B
Llama-3.2-3B-Instruct	meta-llama	Llama 3.2 terms	3.213B
Phi-4-mini-instruct	microsoft	MIT	3.836B
Qwen2.5-1.5B-Instruct	Qwen	Apache-2.0	1.544B
Qwen3.5-4B	Qwen	Apache-2.0	4.660B
Llama-3.2-1B-Instruct	meta-llama	Llama 3.2 terms	1.236B
Mistral-7B-Instruct-v0.3	mistralai	Apache-2.0	7.248B
HarmBench-Mistral-7b-val-cls	cais	MIT	7.242B
Llama-Guard-3-8B	meta-llama	Llama 3.1 terms	8.030B
granite-guardian-3.1-8b	ibm-granite	Apache-2.0	8.171B
Qwen3Guard-Gen-0.6B	Qwen	Apache-2.0	0.752B
Llama-Prompt-Guard-2-86M	meta-llama	Custom; see source	0.279B
shieldgemma-2b	google	Gemma terms	2.614B
wildguard	allenai	Apache-2.0	7.248B
HaloGuard1-Gen-0.8B	astroware	Apache-2.0	1.107B

Table 25: Pretrained encoders, victims, proposers and guards. Source names link to the original publishers. Terms are read from public repository metadata; custom conditions remain governed by the linked release. Parameter totals are reported when exposed in safetensors metadata. They include all stored model parameters, not only a trained readout.

Setting	Qwen reference study	Broad-family study
Optimization	AdamW	AdamW
Updates	1000	2000
Batch size	256 per population	256
Learning rate	0.001	0.001
Weight decay	0.0001	0.0001
Head seeds	0, 1, 2	0, 1, 2
MLP width / dropout	512 / 0.1	512 / 0.1
Gaussian coordinate SD	0.1	Disabled
Positive references	1200 general + 1200 sensitive	8,000 unharmful
Calibration size	600	2,077
Calibration quantile	0.05	0.05
Inspected length	256 tokens	256 tokens

Table 26: Training and calibration settings. The deployed Qwen head is the seed-zero linear arm; other Qwen arms vary references or readout depth. Broad-family recipes use the fixed WildGuardMix references and development selection in Appendix J. Model evaluation uses frozen heads and thresholds.

Package (linked project)	Version
Python	3.11.16
torch	2.13.0
transformers	5.17.0
datasets	5.0.1
accelerate	1.15.0
numpy	2.3.5
pandas	3.0.5
scikit-learn	1.9.1
huggingface-hub	1.31.0
tokenizers	0.23.2
safetensors	0.8.0
peft	0.21.0

Table 27: Verified environment for the native search replay and activation analysis. CUDA 13.0 accompanies the listed PyTorch build. These versions support reproduction; they do not describe every historical experiment.